RUBINA REKIĆ

rrekic1@etf.unsa.ba · +387 60 320 9897 · rubina-rekic.github.io · github.com/rubina-rekic

EDUCATION

University of Sarajevo 2023 - 2026
Bachelor of Engineering in Computer Science *Sarajevo, Bosnia and Herzegovina*
GPA: 8.7/10.0
Thesis: "Application of Graph Theory to Neural Network Modelling"

JU "Gimnazija" Cazin 2019 – 2023
High School Diploma *Cazin, Bosnia and Herzegovina*
Valedictorian

SKILLS

Programming: C#, Kotlin, Java, JavaScript, Python, C++, C, SQL, Git, LaTeX
Web & Frameworks: React, Node.js, Express.js, ASP.NET Core, Flask, Tailwind CSS
Mobile: Android Studio, REST API integration
Other: MQTT, OOP, Design Patterns, TDD
Languages: Bosnian (native) | English (professional) | German (limited working proficiency)

EXPERIENCE AND PROJECTS

Encode d.o.o. March 2025 – Present
Mathematics Instructor
Delivered preparatory classes for university students from the Faculty of Mechanical and Electrical Engineering

Application of Graph Theory to Neural Network Modelling - Bachelor's Thesis 2026
Formalized trained feedforward neural networks as directed weighted graphs and applied centrality, modularity (Louvain), spectral (Laplacian), and robustness analysis to compare three architectures of varying depth and width

ScenarijPro - Full-Stack Web Application 2024/2025
Built a collaborative screenplay editor with real-time conflict resolution, version control checkpoints, and script analysis
Developed iteratively across 4 sprints: from static UI to a full Node.js/Express REST API with MySQL and Sequelize ORM

YourRide - Full-Stack Web Application Spring 2025
Developed a ride-sharing platform using ASP.NET Core with SignalR for real-time notifications and role-based access control
Designed full system architecture including UML diagrams, applied SOLID principles and structural design patterns

FilmMate QA - .NET Application Spring 2025
Applied TDD (Red-Green-Refactor), MSTest unit testing with 90%+ coverage, and white/black-box testing techniques

NewsFeed Android App - Mobile Application Spring 2025
Developed an Android application in Kotlin with real-time news fetching via REST API and local persistence using Room database
Implemented category filtering, sorting, and automatic feed refresh with 3-second polling interval

Neural Networks as Graphs - Seminar Paper 2025

Modeled biological neural networks as directed graphs using graph theory (Dale's Law, Barabási-Albert model) and analyzed their structure and robustness

Laid the groundwork for my Bachelor's thesis, which extended this graph-theoretic approach to ANN